

Abbott 1

Ahmad Rifaie

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1 Introduction

Exercise: Assume $h : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $\mathbb{R}$. Let $K = \{x : h(x) = 0\}$. Show that K is a closed set.

Recall: For $K \subseteq \mathbb{R}$, K is closed $\iff \forall (x_n) \subseteq K$ with $(x_n) \rightarrow x$, we have $x \in K$.

Proof. Let (x_n) be a sequence in K where $(x_n) \rightarrow x$.

Since $x_n \in K$ for all $n \in \mathbb{N}$, we have $h(x_n) = 0$ for all $n \in \mathbb{N}$. Thus,

$$\lim_{n \rightarrow \infty} h(x_n) = 0.$$

Because h is continuous on $\mathbb{R}$, it preserves limits:

$$\lim_{n \rightarrow \infty} h(x_n) = h\left(\lim_{n \rightarrow \infty} x_n\right) = h(x).$$

Equating the limits gives $h(x) = 0$, which implies $x \in K$. □